

PROJECT TITLE

IT4V43 - Machine Learning Operations (MLOps)

Student Placement Prediction System

Presenting By

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Problem Identification and Title Justification

- **Relevance of the problem:**

Student placement is a critical metric for colleges. Currently, institutions struggle to proactively identify students who are at risk of not getting placed until it is too late to intervene. Early identification using data-driven techniques can significantly improve placement outcomes.

- **Clarity of problem statement :**

- Colleges lack an automated, intelligent system to assess and predict the placement likelihood of individual students based on their academic and extracurricular profiles. Manual evaluation is time-consuming, inconsistent, and reactive rather than proactive.

Justification of project title and scope of the project :

The title Student Placement Prediction System precisely captures the core objective — predicting whether a student will be placed — using MLOps principles to ensure the model is production-ready, maintainable, and continuously improving. The scope includes data ingestion, model training, deployment, monitoring, and automated retraining.

Problem Identification and Title Justification

Basic concepts related to the project:

- **MLOps:** Practices combining ML, DevOps, and data engineering for reliable ML lifecycle management
- **Classification Models:** Supervised learning algorithms (Logistic Regression, Random Forest, XGBoost) for binary placement prediction
- **Feature Engineering:** Transforming raw student data (CGPA, Skills, Projects, Internships) into meaningful model inputs
- **Model Drift & Retraining:** Monitoring model accuracy over time and triggering retraining pipelines when performance degrades
- **CI/CD for ML:** Continuous integration and deployment pipelines tailored for ML model updates

Literature Survey

1. Student Placement Prediction using ML

Advantage: Early intervention

Disadvantage: Data dependency

2. Academic Performance Analytics

Advantage: Scalable

Disadvantage: Limited features

3. Placement Intelligence Dashboard

Advantage: Visualization

Disadvantage: Historical bias

4. Employability Prediction

Advantage: Personalized insights

Disadvantage: Retraining required

5. Student Success Analytics

Advantage: Better planning

Disadvantage: Data collection effort

Research Gap:

Combine academics + skills + projects + internships.

Objectives of the Project

- Develop a machine learning model to predict student placement probability based on CGPA, Skills, Projects, and Internships
- Build a **Student Risk Dashboard** to help placement coordinators identify at-risk students early
- Implement an end-to-end **MLOps pipeline** covering data ingestion → training → evaluation → deployment → monitoring
- Design an automated **retraining pipeline** triggered by model performance degradation
- Provide **placement probability scores** per student with explainability (feature importance)
- Ensure the system is scalable, maintainable, and production-ready following MLOps best practices

Project Planning and Time Schedule

- **Gantt chart / timeline preparation:**

Phase	Tasks	Duration	Milestone
Week 1–2	Problem definition, dataset collection, EDA	2 weeks	Dataset ready
Week 3–4	Feature engineering, model selection & training	2 weeks	Baseline model
Week 5–6	MLOps pipeline setup (MLflow, DVC, CI/CD)	2 weeks	Pipeline deployed
Week 7–8	Dashboard development, monitoring integration	2 weeks	Dashboard live
Week 9–10	Retraining pipeline, testing & evaluation	2 weeks	Auto-retrain working
Week 11–12	Final integration, documentation, review	2 weeks	Final submission

Task Allocation Among Team Members:

- **Team Member 1:** Data preprocessing, model training, MLOps pipeline (MLflow, DVC)
- **Team Member 2:** Dashboard development, API integration, retraining pipeline & deployment

References

1. Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.
2. Sculley, D. et al. (2015). Hidden Technical Debt in Machine Learning Systems. NeurIPS.
3. Shankar, V. et al. (2022). Operationalizing Machine Learning: An Interview Study. arXiv:2209.09125.
4. Kumar, R. & Sharma, P. (2021). Student Placement Prediction using Machine Learning Techniques. IJCSE, Vol. 9(4).
5. Raj, M. et al. (2023). An Intelligent Resume-Based Placement Recommender Using NLP and ML. IEEE ICCMC.
6. MLflow Documentation — <https://mlflow.org/docs/latest/index.html>
7. DVC Documentation — <https://dvc.org/doc>

Thank You!